Kartavya Mandora

MS in Artificial Intelligence, Long Island University Brooklyn
kartavyamandora648@gmail.com | (+1) 267-237-8596 | linkedin.com/in/kartavya648 | github.com/Kartavya1905

SUMMARY

I am a master's student in Artificial Intelligence at LIU Brooklyn, with a B.Tech in engineering and experience across software, research, and industry. I am currently looking for opportunities to advance my career in AI and software engineering.

I have built applied ML projects and am currently deepening my knowledge of AI. I also have theoretical knowledge of collective communication in scale-up and scale-out domains.

EDUCATION

Long Island University Brooklyn, MS in Artificial Intelligence (GPA: 3.89/4.00) 2025 - Present

SVNIT Surat, B.Tech, Engineering (GPA: 7.92/10.00) 2020 - 2024

EXPERIENCE

Infinite Civil Solutions Pvt. Ltd., Ahmedabad, India | *Industrial Intern, Pavement Design and Traffic* Jan 2024 - Apr 2024

Pavement Design, IITPAVE, KGPBACK, MS Excel, LCMS RoadInspect, Dynatest Explorer, Traffic Data Analysis, Regression Analysis

- Analyzed national highway traffic and pavement data, building regression-based traffic growth forecasts from GDP/NSDP data and back-calculating structural parameters from FWD deflection measurements to design flexible and rigid pavements.

SVNIT Surat, Water Resources Department | *Research Intern* May 2023 - Jul 2023

Python, HEC-RAS, ArcGIS, ArcHydro, Flood Analysis, Hydraulic Modeling, Land Use Land Cover Classification, Drainage Density Analysis

- Built a land use/land cover classification map of the Tapi Basin from sampled training areas in ArcGIS, and modeled flood risk on the Mahi River with a 1D steady-flow HEC-RAS simulation across 100 km of cross sections.

Acmegrade | *Android App Development Intern* Mar 2023 - May 2023

Android Studio, Java, Android Emulator, App Interface Design

- Completed hands-on Android Studio training, then designed and built a Java shopping app interface that lists a platform's products.
- Built a quiz app in Java where users enter their name, answer questions, and receive a score based on the number of correct answers.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, SQL
- **CS Fundamentals:** Data Structures & Algorithms (practiced in Python and C++ on LeetCode), Computer Networks, System Design, Ethical Hacking
- **AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, CNN, Vision Transformers (ViT), LSTM, BiLSTM
- **Generative AI & LLMs:** Stable Diffusion, ControlNet, Sentence Transformers
- **Information Retrieval:** FAISS, Semantic Search, Vector Embeddings, RAG Pipelines, PDF Processing
- **Computer Vision & Audio:** OpenCV, Albumentations, Image Preprocessing, Audio Signal Processing
- **Frameworks & Developer Tools:** Flask, Streamlit, Gradio, Git, Docker, Jupyter Notebook, Google Colab
- **Theoretical Knowledge:** Scale-Out Networking Protocols like RoCEv2 and RDMA, Collective Communications (AllReduce, AllGather), Data, Pipeline, and Tensor Parallelism, PFC, ECN, DCQCN

PROJECTS

Sketch2Style | Stable Diffusion, ControlNet *Ongoing Research*

- Turning hand-drawn room sketches into photorealistic interior designs with Stable Diffusion and ControlNet, plus custom preprocessing.
- Investigating evaluation methods measuring structural fidelity, visual realism, and user-guided consistency for an ongoing publication.

Research Paper Recommender | Sentence-BERT, FAISS

- Built a paper recommender with PDF text extraction, section-aware processing, Sentence-BERT embeddings, and FAISS search.
- Designed a human-in-the-loop workflow with confidence thresholding, feedback collection, and citation export for reliable retrieval.
- Tested each pipeline step on real papers and measured match quality on a small test set of papers with known related work.

Skin Cancer Classification using Vision Transformers | PyTorch, Albumentations

- Built a multi-class skin lesion classifier on HAM10000 in PyTorch with augmentation, stratified splits, and transfer learning.
- An early CNN overfit because one class dominated the data; moving to a Vision Transformer and per-class metrics exposed it.

DNA Splice Site Classification using CNN-BiLSTM | TensorFlow

- Built a hybrid CNN-BiLSTM in TensorFlow on 45,000+ DNA sequences with one-hot encoding, stratified splits, and class balancing.
- Achieved 90.5% test accuracy and outperformed SVM and Random Forest baselines on ROC-AUC, Precision-Recall, and F1-score.

Elderly Voice Recognition using Deep Learning | Audio Signal Processing

- Built a speech classifier for elderly voices using noise reduction, normalization, segmentation, and acoustic feature engineering.
- Evaluated with precision, recall, F1-score, and confusion matrices, not accuracy alone.

Neural Network from Scratch | Pure Python, No Libraries *In Progress*

- Writing a neural network in plain Python with zero imports, with custom exp, log, sqrt, and sigmoid built from Taylor series.
- Matched Python's math library to about 13 decimal places and got a working XOR forward pass; backpropagation is next.

ACHIEVEMENTS

- Finalist, USAII Global AI Hackathon 2026 (Graduate Level), for the Research Paper Recommender, built as a two-member team.
- Secured 96.27 percentile in the Joint Entrance Examination (JEE) Main, India.